

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Requirements

Online Complaint Registration and Management Systems — MERN
Stack E-commerce Application

Phase: Requirement Analysis Phase

Document Version: 1.1

Date: 18th of june, 2026

Prepared by: BHAVINI SURI

Classification: Project Documentation — For Academic Submission

Solution Requirements

1. Introduction

This document specifies the complete set of Functional and Non-Functional Requirements for the Online Complaint Registration and Management System (OCRMS). Each requirement is assigned a unique identifier to enable traceability from requirement analysis through design, implementation, testing, and deployment.

2. Functional Requirements

FR-01 to FR-05 – User Management

Req. ID	Requirement Description	Priority	USN
FR-01	The system shall permit new users to register by providing a unique email address, full name, and password. Email must be validated for format correctness.	Must Have	USN-1
FR-02	The system shall authenticate registered users via email and password credentials and issue a signed JWT token upon success.	Must Have	USN-2
FR-03	The system shall store JWT tokens exclusively in HTTP-only cookies to prevent client-side access and improve security	Must Have	USN-2
FR-04	The system shall enforce Role-Based Access Control (RBAC) distinguishing between the <code>user</code> and <code>admin</code> roles.	Must Have	USN-0
FR-05	Authenticated users shall be able to update their profile including name, password, phone number, and contact information	Should Have	USN-1

FR-06 to FR-10 – Product Catalogue

Req. ID	Requirement Description	Priority	USN
FR-06	The system shall allow citizens to submit complaints by selecting a category and providing detailed complaint information	Must Have	USN-3
FR-07	The system shall allow users to search and track complaints using complaint ID and status information.	Must Have	USN-3
FR-08	The system shall support complaint filtering by category, status, department, and date	Must Have	USN-3
FR-09	The system shall provide complaint sorting based on date submitted, priority, and status.	Should Have	USN-3
FR-10	Administrators shall be able to create, assign, update, and manage complaint records and officer assignments.	Must Have	USN-8

FR-11 to FR-15 – Shopping Cart & Checkout

Req. ID	Requirement Description	Priority	USN
FR-11	The system shall allow officers to manage assigned complaints and update complaint statuses	Must Have	USN-4
FR-12	The system shall allow users and officers to communicate through real-time chat functionality.	Must Have	USN-4
FR-13	The system shall provide a complaint workflow consisting of Registration → Assignment → Investigation → Resolution → Closure	Must Have	USN-5
FR-14	The system shall generate real-time notifications for complaint assignments, status updates, and resolutions	Must Have	USN-6
FR-15	Upon complaint resolution, the system shall allow citizens to submit ratings and feedback for service evaluation.	Must Have	USN-6

3. Non-Functional Requirements

Req. ID	Category	Requirement Description	Priority
NFR-01	Security	All user passwords must be hashed using Bcrypt (salt rounds ≥ 10) prior to persistence.	Must Have
NFR-02	Security	All API routes that modify data must require a valid JWT; unauthenticated requests shall return HTTP 401.	Must Have
NFR-03	Security	Administrative routes shall enforce role-based authorization and return HTTP 403 for unauthorized access	Must Have
NFR-04	Performance	Complaint retrieval and status update APIs must respond within 500ms under normal load conditions.	Should Have
NFR-05	Performance	Real-time notifications and chat messages must be delivered with minimal latency.	Should Have
NFR-06	Usability	The frontend must be fully responsive across mobile, tablet, and desktop devices.	Must Have
NFR-07	Reliability	MongoDB transactions and validation mechanisms must ensure data consistency during complaint processing.	Must Have

Req. ID	Category	Requirement Description	Priority
NFR-08	Maintainability	Backend code must follow the MVC architecture pattern with clear separation of concerns.	Should Have
NFR-09	Scalability	The stateless JWT architecture shall support horizontal scaling without shared session state.	Could Have

4. Requirements Traceability Matrix

Req. ID	Design Artefact	Implementation File	UAT Test Case
FR-01, FR-02, FR-03	Solution Architecture	userController.js	UAT-001, UAT-002
FR-06, FR-07, FR-08	Data Flow Diagram	ComplaintController.js	UAT-005, UAT-006
FR-11, FR-12	ER Diagram	complaint.js model	UAT-008, UAT-009
FR-13, FR-14, FR-15	User Flow Sequence	notificationController.js	UAT-011, UAT-012
FR-10	Solution Architecture	adminController.js	UAT-013
NFR-01, NFR-02	Solution Architecture	authMiddleware.js	UAT-004, UAT-016

5. Conclusion

The Solution Requirements document defines the functional and non-functional requirements necessary for developing OCRMS. These requirements establish a clear foundation for system design, implementation, testing, and deployment while ensuring security, scalability, reliability, and effective complaint management.